

Total No. of Questions : 8]

SEAT No. :

P-7545

[Total No. of Pages : 3

[6180]-53

T.E. (Computer Engineering)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6. Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific calculator is permitted.

- Q1)** a) Explain Data Analytics Cycle with suitable diagram and its phases. [8]
b) List and Explain the various activities involved in identifying potential data resources as a part of discovery phase in Data Analytics Life Cycle? [9]

OR

- Q2)** a) List and explain the key roles for successful analytics project. [8]
b) Write short note on : [9]
i) Common Tools for the Model Building
ii) Model selection for Data Analytics
- Q3)** a) List and explain the various types of analytics in Big data. [9]
b) Calculates the support and confidence value for all the possible item sets. [9]

Transaction ID	Items bought
1	Onion, Potato, Cold Drink
2	Onion, Burger, Cold Drink
3	Eggs, Onion, Cold Drink
4	Potato, Milk, Eggs
5	Potato, Burger, Cold Drink, Milk, Eggs

OR

P.T.O.

- Q4) a)** Explain the need of logistic regression along with its various types. [9]
- b)** Explain the following terms with suitable example. [9]
- Removing Duplicates from dataset.
 - Handling Missing Data

- Q5) a)** Suppose that the given data the task is to cluster points (with (x, y) representing location) into three clusters, where the points are A1 (2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Euclidean distance. Suppose initially we assign A1, B1 and C1 as the center of each cluster, respectively. [8]

Use the k-means algorithm to show only show only the first round of execution with cluster center.

- b)** Explain the following Text Analysis steps with suitable example [9]
- Part-of-speech(POS)tagging
 - Lemmatization

OR

- Q6) a)** Given the confusion matrix, Calculate Accuracy, Precision, Recall, Error rate with description on Diabetic Risk. [8]

	Classes	Predicted classes	
		Diabetic Risk -Yes	Diabetic Risk -No
Actual classes	Diabetic Risk- Yes	90	210
	Diabetic Risk- No	140	9560

- b)** Explain the Text Preprocessing steps with suitable example. [9]

- Q7) a)** List the few data visualization tools and discuss any four applications of data visualization along with the use of the various plots with Python/R or suitable tool. [9]
- b)** List the challenges of Data Visualization. Explain the types of visualization with example. [9]

OR

- Q8)** a) Explain in detail the Hadoop Ecosystem with suitable diagram along with the various components. [9]
- b) Write a short note on the following. [9]
- a) Map Reduce
- b) Pig



Q1)

a) **Data Analytics Cycle and Phases:**

The **Data Analytics Cycle** is a structured approach to process and analyze data in a way that leads to actionable insights. It involves several phases, each designed to handle specific tasks and challenges related to data analysis. Below is an explanation of the cycle with its phases:

Phases of the Data Analytics Cycle:

1. **Data Collection:** This is the first phase where data is gathered from various sources, such as databases, APIs, spreadsheets, or real-time data streams. The objective is to ensure that the data is relevant to the problem at hand.
2. **Data Cleaning:** In this phase, the collected data is cleaned and pre-processed. This involves handling missing values, eliminating duplicates, dealing with outliers, and transforming the data into a format that is suitable for analysis.
3. **Data Exploration:** In this phase, exploratory data analysis (EDA) is performed. Analysts visualize and examine the data to discover patterns, trends, and relationships among the variables.

This helps in forming hypotheses and understanding the data's underlying structure.

4. ****Data Modeling****: The data is then used to build predictive models. This involves selecting appropriate algorithms, training the models on the data, and validating their accuracy. The goal is to derive insights or make predictions based on the data.

5. ****Evaluation and Interpretation****: In this phase, the performance of the models is evaluated using various metrics like accuracy, precision, recall, etc. Insights from the data are interpreted to understand the significance of the findings.

6. ****Deployment and Monitoring****: Finally, the model is deployed to the real-world environment where it will make decisions or predictions. The model is continuously monitored to ensure it performs well and is adjusted as needed.

****Data Analytics Cycle Diagram:****

```
+-----+ +-----+ +-----+
| Data | -> | Data Cleaning | -> | Data Exploration |
| Collection | | & Preprocessing | | & Visualization |
```

```
+-----+ +-----+ +-----+
```

|||

v v v

+-----+ +-----+ +-----+

| Data Modeling | -> | Evaluation & | -> | Deployment & |

| (Algorithm | | Interpretation | | Monitoring |

| Building) | +-----+ +-----+ +-----+

b) ****Identifying Potential Data Resources in the Discovery Phase:****

The discovery phase in the ****Data Analytics Life Cycle**** is focused on identifying, collecting, and validating data resources that will be used for analysis. It involves several activities that ensure the necessary data is available and relevant for the problem at hand. Some of these activities are:

1. ****Understanding Business Objectives****: The first activity is to align the data resources with the business goals. This involves discussions with stakeholders to understand what questions need to be answered and how data can support those objectives.

2. ****Identifying Relevant Data Sources****: This involves exploring both internal and external data sources that might contain the relevant information. Internal sources include databases, logs, and previous reports, while external sources may include public datasets, APIs, or third-party data providers.

3. ****Data Source Evaluation****: Once potential data sources are identified, it's important to evaluate their quality, relevance, and completeness. This includes checking the data's accuracy, consistency, and availability.
4. ****Data Acquisition****: This activity involves obtaining the data, which could mean importing data from various systems, web scraping, connecting to APIs, or acquiring third-party data.
5. ****Ensuring Data Accessibility****: The selected data must be accessible and usable. This may involve setting up systems for data integration or transformation, ensuring that the data can be processed easily.
6. ****Data Governance and Compliance****: In this step, any legal or regulatory considerations regarding data usage must be addressed. This includes ensuring that the data is being used in accordance with privacy laws, such as GDPR or HIPAA.
7. ****Data Quality Assessment****: Finally, the quality of the data is assessed, including checking for missing values, duplicates, and anomalies. Ensuring high-quality data is crucial for the accuracy of the analysis.

a) ****Key Roles for a Successful Analytics Project:****

For a data analytics project to be successful, various roles need to be fulfilled, with each role focusing on specific areas of the project lifecycle. Some key roles are:

1. ****Data Scientist****: The data scientist is responsible for applying mathematical and statistical techniques to analyze data and build models. They typically work on designing experiments, developing algorithms, and interpreting results.

2. ****Data Engineer****: The data engineer is responsible for managing, preparing, and transforming data for analysis. They build and maintain the data pipelines and ensure the smooth flow of data from different sources to the analysis environment.

3. ****Business Analyst****: The business analyst works closely with stakeholders to understand business goals and requirements. They translate business problems into data questions and ensure that the analytics team addresses the right objectives.

4. ****Data Analyst****: The data analyst works with the data and performs exploratory analysis. They produce reports, dashboards, and visualizations to provide insights into the data.

5. **Project Manager**: The project manager ensures that the project is completed on time, within scope, and within budget. They handle timelines, resources, and communication between teams.

6. **Domain Expert**: The domain expert provides deep knowledge of the business or industry to guide the analytics team in understanding the context of the data and its implications for the business.

7. **IT/Infrastructure Specialist**: This role is responsible for managing the technical infrastructure, such as cloud storage, servers, and databases, ensuring that all tools and platforms used in the project run smoothly.

b) **Short Notes on:**

i) **Common Tools for Model Building:**

Some common tools used for model building in data analytics and machine learning include:

- **Python Libraries**: Libraries like Pandas (data manipulation), Scikit-learn (machine learning algorithms), TensorFlow, and PyTorch (deep learning) are commonly used for building models.
- **R**: R is another powerful language for statistical analysis

and model building, particularly in academic research.

- **Jupyter Notebooks**: A popular environment for building, documenting, and sharing data analysis and model-building code.
- **Big Data Platforms**: Tools like Apache Spark and Hadoop can be used for processing large datasets and building scalable models.
- **Data Visualization Tools**: Tools like Tableau, Power BI, or matplotlib (Python) are often used for visualizing the results of models.

ii) **Model Selection for Data Analytics**

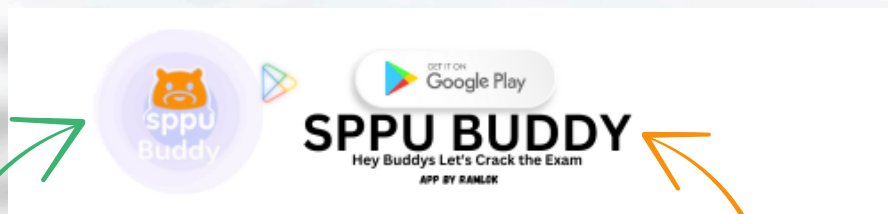
Model selection is a critical step in the data analytics process. It involves choosing the right algorithm or model that best fits the data and business problem. Key factors in model selection include:

- **Type of Problem**: Whether the problem is regression, classification, clustering, or recommendation, different models are suited to different types of problems. For instance, linear regression is good for regression problems, while decision trees or neural networks might be better for classification problems.
- **Data Size**: Some models, such as decision trees and logistic regression, perform well with smaller datasets, while others like neural networks require large datasets to work effectively.

- ****Accuracy vs. Interpretability****: Some models (like linear regression) are easy to interpret, while others (like deep neural networks) offer high accuracy but lack transparency. Depending on the project's needs, one may prioritize accuracy or interpretability.
- ****Computational Resources****: Some models, such as deep learning, require significant computational power, while simpler models like linear regression are less resource-intensive.
- ****Evaluation Metrics****: The choice of model is also influenced by the evaluation metrics used, such as accuracy, precision, recall, or F1-score, depending on the specific goals of the analysis.



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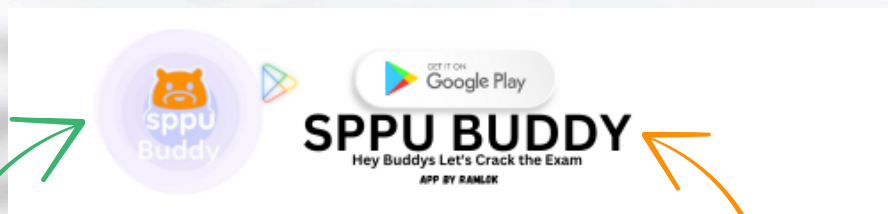


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